

# Roberto Cárdenas Retamal

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*Last updated: October 2026*

## EDUCATION

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2022 – Aug 2026      **Ph.D. in Fisheries and Aquatic Sciences**, University of Florida

*Dissertation:* “Incentives and Attributes in Seafood Markets”

*Co-Chairs:* Dr. Frank Asche, Dr. Di Fang

2016 – 2018      **MSc in Natural and Resources Economics**, Universidad de Concepción

*Advisor:* Dr. Jorge Dresdner

2011 – 2015      **BS in Economics**, Universidad de Concepción

## ACADEMIC APPOINTMENTS

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Aug 2026 – Present      **Research Associate (Economic Research)**, University of Florida

## FIELDS

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Food Policy and Nutrition Economics, Applied Econometrics (Causal Inference, Machine Learning), Environmental and Natural Resource Economics

## JOB MARKET PAPER

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“Seafood Purchasing in Response to Type 2 Diabetes Diagnosis,” with Di Fang and Frank Asche  
Using NielsenIQ Consumer Panel and Ailment Survey data, this paper identifies newly diagnosed Type 2 diabetic households with a random forest classifier and estimates a probability-weighted event-study regression showing that diagnosis triggers a persistent increase in seafood purchases (5.8% on impact, 14.6% in the following months), concentrated among larger, older, and lower- to middle-income households. *Status: Under review at Food Policy.*

## PAPERS UNDER REVIEW

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1. “Seafood Purchasing in Response to Type 2 Diabetes Diagnosis” with Di Fang and Frank Asche  
*Status: Under review at Food Policy*
2. “Market Penetration and Economic Potential of Crawfish, Oysters, and Red Drum in Southeastern U.S. Markets” with Quinn LaFontaine, Taryn Garlock, Frank Asche, Andrew Ropicki, and Bixuan Yang  
*Status: Submitted to Aquaculture*

## ACADEMIC PUBLICATIONS

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Total Citations: 166 | H-index: 6

1. Liu, J., Li, T., Patton, M., **Cardenas-Retamal, R.**, Roy, D. (2026). “More than just a break: performance after walking in green and urban spaces.” *Applied Economics*.

2. Iversen, A., Asche, F., Nyrud, T., Robertsen, R., Bendiksen, B.I., Steinsbø, S. and **Cardenas-Retamal, R.** (2025). “The growing importance of aquaculture: The case of Norwegian salmon.” *Reviews in Aquaculture*, 17(4): e70088.
3. Ceballos-Concha, A., Asche, F., **Cardenas-Retamal, R.** (2025). “Salmon aquaculture in Chile: Production growth and socio-economic impacts.” *Reviews in Aquaculture*, 17(1): e12993.
4. Salazar, C., Jaime, M., **Cárdenas, R.**, Hernández, F. (2025). “Risk perception and production risk in seaweed aquaculture.” *Aquaculture Economics & Management*, 1–29.
5. Salazar, C., Jaime, M., **Cardenas Retamal, R.**, Baquedano, M. (2023). “Women engagement, psychological traits, and gender gaps in the small-scale seaweed aquaculture in Chile.” *Reviews in Aquaculture*, 15(4): 1540–1553.
6. Salazar, C., **Cardenas Retamal, R.**, Jaime, M. (2023). “Environmental efficiency in the salmon industry—an exploratory analysis around the 2007 ISA virus outbreak and subsequent regulations in Chile.” *Environment, Development and Sustainability*, 25: 8107–8135.
7. **Cardenas Retamal, R.**, Barrientos, M., Vasquez-Lavin, F., Ponce, R. (2023). “Intrahousehold bargaining power and time allocation for multiple activities.” *Latin American Economic Review*, 32: 1–22.
8. **Cardenas Retamal, R.**, Dresdner-Cid, J., Ceballos-Concha, A. (2021). “Impact assessment of salmon farming on income distribution in remote coastal areas: The Chilean case.” *Food Policy*, 101: 102078.

## EXTENSION PUBLICATIONS

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1. “What New and Repositioned Fruits Teach Us: Lessons for the U.S. Muscadine Market,” with Ming Ge, Di Fang, Melinda Knuth, and Lisa House. *Status: In preparation; target outlet: Applied Economics & Education: Applied Economic Insights*.
2. Ceballos, A., Cardenas Retamal, R., Dresdner, J. (2023). “Evaluación del impacto socioeconómico de la industria del salmón. Pobreza y distribución del ingreso en zonas costeras rurales en Los Lagos, Chile.” [“Socioeconomic impact assessment of the salmon industry: Poverty and income distribution in rural coastal areas in Los Lagos, Chile.”] (in Spanish). *Interdisciplinary Center for Aquaculture Research (INCAR)*, Policy Brief N° 13.

## WORK IN PROGRESS

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- “Price premiums for the MSC-ecolabel in the U.S.: A hedonic analysis of haddock in the U.S. market,” with Frank Asche, Di Fang, and Adams Ceballos
- “The role of trust on the willingness to pay for enhanced Harmful Algal Blooms policies,” with Felipe Vasquez, Manuel Barrientos, Roberto Ponce, and Stefan Gelcich
- “Pregnancy and the Seafood Demand Response: Evidence from Household Scanner Data,” with Frank Asche, Di Fang, Jordan Moor, and Thomas Anderson
- “The Value of Nutritional Attributes on Seafood,” with Frank Asche, Di Fang, and Jordan Moor
- “Harmful Algal Blooms and Fishing Choices: Understanding Stone Crabbers’ Responses to Ecological Disruptions,” with Adams Ceballos and Jordan Moor

- “Fishing Location Choice under Environmental and Regulatory Constraints: Evidence from Southern Chile”
- “Harmful Algal Blooms and Coastal Employment: Causal Evidence from Chile”
- “Vertical Differentiation and Farm Location in the Salmon Aquaculture Industry,” with Thomas Anderson
- “Not All Protein Is Equal: Seafood in the Demand for Animal Protein”

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## NON-AUTHORED PAPERS I SUPPORTED WITH RESEARCH ASSISTANCE

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1. Branco, D., Carrillo, B., Fang, D., & Iglesias, W. (2025). “Expertise Overlap and Team Productivity: Evidence from the Hospital Industry.” *Journal of Political Economy: Microeconomics*.

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## RESEARCH EXPERIENCE

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Research Associate (Economic Research) — University of Florida Aug 2026 – Present

- Designed producer and buyer survey instruments to measure supply and demand in the Florida native seed market, part of a broader economic analysis framework covering market structure, price formation, and forecasting scenarios for a market with limited price discovery and multi-year supply lags between planting and harvest.

Graduate Research Assistant — Fisheries and Aquatic Sciences, University of Florida 2022 – Aug 2026

- Designed and executed three dissertation studies using NielsenIQ household scanner data to test how market information, health diagnoses, and life-stage transitions change food purchasing behavior.
- Job market paper: identified newly diagnosed Type 2 diabetic households with a random forest classifier and estimated a probability-weighted regression on NielsenIQ Consumer Panel data linked to an annual health survey; found diagnosis raises seafood purchases by about 5.8% on impact and 14.6% in the following months, concentrated among larger, older, and lower- to middle-income households, with the largest gains among previously low-consuming households and direct relevance to the 2024 addition of canned seafood to the WIC food package (with Frank Asche and Di Fang; under review, *Food Policy*).
- Used ten years of NielsenIQ scanner data (2010–2020) and a difference-in-differences design to show that pregnancy triggers a persistent, dynamically deepening reduction in seafood purchases, reaching 4.7 percentage points two years post-conception across both high- and low-mercury species, contrary to FDA advisory guidance; extended the analysis with a causal forest identifying household size, income, and pre-pregnancy purchase intensity, rather than age, as the substantive drivers of heterogeneity.
- Built a hedonic pricing model estimating the first U.S. market price premium for Marine Stewardship Council (MSC) sustainability-certified haddock (manuscript under review, *Journal of the Agricultural and Applied Economics Association*).
- Investigating consumer valuation of seafood’s nutritional attributes, including omega-3 (DHA/EPA) content, with Frank Asche, Di Fang, and Jordan Moor (in progress).
- Contributed to the literature review for “Expertise Overlap and Team Productivity: Evidence from the Hospital Industry” (Branco, Carrillo, Fang, and Iglesias, 2025), published in the *Journal of*

*Political Economy: Microeconomics.*

- Contributed statistical and analytic sections to a successful \$525,404 USDA NIFA proposal, “Understanding Dietary Change and Food Purchase for Better Policy Design” (Grant No. 2026-67023-45958), led by Dr. Di Fang, participating directly in the writing of the proposal that was ultimately funded.

Research Assistant — Africa’s Seafood Trade and Nutrition Project, University of Florida 2025 – 2026

- Built and analyzed a 25-year bilateral trade panel (3.15M records, 54 countries, 1,759 species) linked to nutrient composition data to quantify cross-border flows of protein, DHA/EPA, iron, zinc, and vitamin B12.
- Estimated structural gravity models with high-dimensional fixed effects and Gini/HHI/Theil inequality decompositions; built a reproducible R analysis pipeline harmonizing product- and country-level nutrient attribute data.

University of Florida (USDA NIFA Specialty Crop Research Initiative) 2025

*Research Assistant: “Through the Grapevine: Developing Vitis x Muscadinia Wide Hybrids for Enhanced Disease Resistance and Quality.”*

Interdisciplinary Center for Aquaculture Research (INCAR), Universidad de Concepción 2019 – 2022

- Worked as a field survey collector, conducting household and community-level surveys as part of primary data collection.
- Conducted applied econometric analysis on aquaculture, household labor allocation, and regional income distribution, contributing to peer-reviewed publications and policy briefs.
- Co-authored a policy brief synthesizing peer-reviewed evidence for a live, multi-stakeholder policy debate, translating technical findings for policymakers, industry, civil society, and media.

Environment for Development (EfD) Initiative – Chile 2022

*Research Assistant.*

Economic Commission for Latin America and the Caribbean (ECLAC) 2020

*Research Assistant.*

Center of Applied Ecology and Sustainability (CAPES) 2020

*Research Assistant.*

## TEACHING EXPERIENCE

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Spring 2025 **Guest Lecturer**, International Agribusiness Marketing (AEB4343), University of Florida

Spring 2023 **Teaching Assistant**, Agricultural Macroeconomics (AEB 3281), University of Florida

Fall 2022 **Teaching Assistant**, Agricultural and Natural Resource Ethics (AEB 4126), University of Florida

2022 **Lecturer**, Statistics for Management; Economics, Sustainability and Entrepreneurship, Universidad Santo Tomás

2018 – 2020 **Lecturer**, Econometrics, Microeconomics, Universidad del Bío-Bío

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|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 – 2022 | <b>Lecturer</b> , Development Economics, Environmental Economics, Universidad de Concepción                                                                                             |
| 2019 – 2021 | <b>Lecturer</b> , Introduction to Economics, Microeconomics, Econometrics, Natural Resources and Environment Economics, Macroeconomics, Universidad Católica de la Santísima Concepción |

## TEACHING FIELDS

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*Undergraduate:* Econometrics, Microeconomics, Macroeconomics, Development Economics, Environmental and Natural Resource Economics, Agricultural Economics

*Graduate:* Applied Econometrics (Causal Inference), Environmental and Natural Resource Economics, Food and Nutrition Economics

## HONORS, AWARDS, AND FELLOWSHIPS

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|      |                                                                                                                                                                                                      |
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| 2025 | Top 10 Most-Cited Articles Published by <i>Reviews in Aquaculture</i> in 2024, for “Salmon aquaculture in Chile: Production growth and socio-economic impacts” (with Adams Ceballos and Frank Asche) |
| 2025 | Travel Scholarship – Summer School in Environmental and Energy Economics, University of California, Berkeley                                                                                         |
| 2022 | Global Food System Institute Fellowship, University of Florida                                                                                                                                       |
| 2022 | Grinter Fellowship Award, University of Florida                                                                                                                                                      |
| 2022 | Latin American and Caribbean (LAC) Scholarship, University of Florida                                                                                                                                |

## GRANTS AND RESEARCH FUNDING

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| 2026 – 2029 | Proposal Contributor, “Understanding Dietary Change and Food Purchase for Better Policy Design,” USDA NIFA (Grant No. 2026-67023-45958, \$525,404; PI: Di Fang), University of Florida |
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## CONFERENCE AND SEMINAR PRESENTATIONS

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\* *presenting author*

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| July 2026 | Cardenas Retamal, R.*; Asche, F.; Fang, D. “The value of nutritional attributes in seafood.” AAEA Annual Meeting. Kansas City, MO. ( <i>Accepted</i> )        |
| July 2026 | Cardenas Retamal, R.*; Asche, F.; Fang, D. “Labor outcomes effects of mental health.” AAEA Annual Meeting. Kansas City, MO. ( <i>Accepted</i> )               |
| Feb. 2026 | Cardenas Retamal, R.* “The value of nutritional attributes in seafood.” Fisheries and Aquatic Sciences Graduate Student Symposium. Gainesville, FL.           |
| Aug. 2025 | Cardenas Retamal, R.* “Demand for environmental attributes in the U.S. seafood market.” Summer School in Environmental and Energy Economics, UC Berkeley. CA. |
| Mar. 2025 | Cardenas Retamal, R.*; Asche, F.; Fang, D. “Sustainable seafood consumption: A hedonic analysis of ecolabeling in the U.S. market.” NAAFE. La Jolla, CA.      |

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| Feb. 2025 | Cardenas Retamal, R.*; Fang, D.; Asche, F. “Disease diagnosis and seafood purchases in the U.S.” Fisheries and Aquatic Sciences Graduate Student Symposium. Gainesville, FL.                                                                           |
| Nov. 2024 | Cardenas Retamal, R.*; Fang, D.; Asche, F. “Seafood purchase, diet-related disease, and WIC participation.” Southern Economic Association (SEA). Washington D.C.                                                                                       |
| 2024      | Ceballos, A.; Asche, F.; Cardenas, R.* “Salmon aquaculture in Chile: Production growth and socio-economic impact.” Aqua2024, World Aquaculture Society. Copenhagen, Denmark.                                                                           |
| 2024      | Cardenas Retamal, R.*; Fang, D.; Asche, F. “Seafood purchase, diet-related disease, and WIC participation.” AAEEA Annual Meeting. New Orleans, LA.                                                                                                     |
| 2023      | Ceballos, A.; Asche, F.; Cardenas, R.* “The socioeconomic impact of salmon aquaculture in Chile.” Fisheries and Aquatic Sciences Graduate Student Symposium, Poster Session. Gainesville, FL.                                                          |
| 2023      | Ceballos, A.; Asche, F.; Cardenas, R.* “The socioeconomic impact of salmon aquaculture in Chile.” Aquaculture America, World Aquaculture Society. New Orleans, LA.                                                                                     |
| 2022      | Cardenas, R.*; Ceballos, A.; Asche, F. “The socioeconomic impact of salmon aquaculture in Chile.” VIII Annual Meeting, Environment for Development Initiative – Chile.                                                                                 |
| 2022      | Cardenas, R.*; Orellana, F.* “A new economy for a new constitution.” Commission on the Environment, Rights of Nature, Natural Commons and Economic Model, Constitutional Convention, Chile.                                                            |
| 2021      | Cardenas Retamal, R.*; Barrientos, M.; Vasquez-Lavin, F.; Ponce, R. “Intra-household bargaining power and time allocation for multiple activities.” XI Annual Meeting of the Chilean Society of Regional Studies (SOCHER).                             |
| 2021      | Salazar, C.; Cardenas Retamal, R.*; Jaime, M. “Environmental efficiency in the salmon industry: An exploratory analysis around the ISA virus and subsequent regulations in Chile.” VII Annual Meeting, Environment for Development Initiative – Chile. |

## PROFESSIONAL SERVICE

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- Referee: *Environment, Development and Sustainability, Frontiers in Human Dynamics, Agricultural and Resources Economics Review, Aquaculture Economics & Management, and Aquaculture International*
  - Member: Agricultural & Applied Economics Association (AAEA), Association of Environmental and Resource Economists (AERE), European Aquaculture Society – World Aquaculture Society, North American Association of Fisheries Economists (NAAFE)
  - Treasurer, Graduate Student Organization (2024–2025)
  - Volunteer, Future of Food Forum 2024, University of Florida

## COMPUTATIONAL SKILLS

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**Statistical & Programming:** R, Stata, Python, SQL, QGIS, LaTeX

**Methods:** Causal inference (difference-in-differences, event-study design, causal forests), hedonic price and demand estimation, discrete choice modeling, nutrient/food attribute data harmonization, study design, assessment of bias and confounding

## LANGUAGES

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Spanish (native), English (fluent)

## REFERENCES

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**Prof. Frank Asche**

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